Assignment-12.5

Perumalla Sushwanth

batch 29

2303a51567

Lab 12.5: Algorithms with AI Assistance – Sorting, Searching, and Optimizing Algorithms.

Task 1 :

(Sorting – Merge Sort Implementation)

• Task: Use AI to generate a Python program that implements the Merge Sort algorithm.

• Instructions:
o Prompt AI to create a function merge_sort(arr) that sorts a list in ascending order.
o Ask AI to include time complexity and space complexity in the function docstring.
o Verify the generated code with test cases.

• Expected Output:
o A functional Python script implementing Merge Sort with proper documentation

PROMPT:

Generate a Python function merge_sort(arr) that sorts a list in ascending order using the Merge Sort algorithm and include time complexity and space complexity

in the function docstring

CODE:

```python
# Generate a Python function merge_sort(arr) that sorts a list in ascending order using the Merge Sort algorithm and include time complexity and space complexity in the function docstring
def merge_sort(arr):
    """
    Sorts a list using Merge Sort algorithm.

    Time Complexity: O(n log n)
    Space Complexity: O(n)
    """

    if len(arr) > 1:
        mid = len(arr) // 2
        left = arr[:mid]
        right = arr[mid:]

        merge_sort(left)
        merge_sort(right)

        i = j = k = 0

        while i < len(left) and j < len(right):
            if left[i] < right[j]:
                arr[k] = left[i]
                i += 1
            else:
                arr[k] = right[j]
                j += 1
            k += 1

        while i < len(left):
            arr[k] = left[i]
            i += 1
            k += 1

        while j < len(right):
            arr[k] = right[j]
            j += 1
            k += 1

    return arr

# Example usage
arr = [38, 27, 43, 3, 9, 82, 10]
print("Sorted Array:", merge_sort(arr))
```

OUTPUT:

```
PS C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING>  & 'C:\Users\perum\AppData\Local\Programs\Python\Python312\python.exe' 'c:\Users\perum\.vscode\extensions\ms-python.
ebugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61819' '--' 'C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING\ai_assistedcoding\lab_12.5'
Sorted Array: [3, 9, 10, 27, 38, 43, 82]
PS C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING>
```

TASK 2:

(Searching – Binary Search with AI Optimization)

- Task: Use AI to create a binary search function that finds a target element in a sorted list.

- Instructions:

o Prompt AI to create a function binary_search(arr, target) returning the index of the target or -1 if not found.
o Include docstrings explaining best, average, and worst-case complexities.
o Test with various inputs.

• Expected Output:
o Python code implementing binary search with AI- generated comments and docstrings.

PROMPT:

Generate a Python function merge_sort(arr) that sorts a list in ascending order using the Merge Sort algorithm and include time complexity and space complexity in the function docstring

CODE:

```python
# Generate a Python function merge_sort(arr) that sorts a list in ascending order using the Merge Sort algorithm and include time complexity and space complexity in the function docstring
#
def merge_sort(arr):
    """
    Sorts a list using Merge Sort algorithm.

    Time Complexity: O(n log n)
    Space Complexity: O(n)
    """

    if len(arr) > 1:
        mid = len(arr) // 2
        left = arr[:mid]
        right = arr[mid:]

        merge_sort(left)
        merge_sort(right)

        i = j = k = 0

        while i < len(left) and j < len(right):
            if left[i] < right[j]:
                arr[k] = left[i]
                i += 1
            else:
                arr[k] = right[j]
                j += 1
            k += 1

        while i < len(left):
            arr[k] = left[i]
            i += 1
            k += 1

        while j < len(right):
            arr[k] = right[j]
            j += 1
            k += 1

    return arr

# Example usage
arr = [38, 27, 43, 3, 9, 82, 10]
print("Sorted Array:", merge_sort(arr))
```

OUPUT:

```
' 'C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING\ai_assistedcoding\lab_12.5'
Sorted Array: [3, 9, 10, 27, 38, 43, 82]
PS C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING>
```

TASK 3:

Smart Healthcare Appointment Scheduling
System
A healthcare platform maintains appointment records containing appointment ID, patient name, doctor name, appointment time, and consultation fee. The system needs to:

1. Search appointments using appointment ID.
2. Sort appointments based on time or consultation fee.
   Student Task
   • Use AI to recommend suitable searching and sorting algorithms.
   • Justify the selected algorithms.
   • Implement the algorithms in Python.

PROMPT:

```
# Here is the **one-line prompt** for Task Description #3:
#"Generate a Python program for a Smart Healthcare Appointmen
t Scheduling System that stores appointment ID, patient name,
doctor name, appointment time, and consultation fee, uses Bin
ary Search to find appointments by ID, and uses Merge Sort to
sort appointments by time or consultation fee with proper com
ments and explanations."**
# Smart Healthcare Appointment Scheduling System
```

CODE:

```python
# Here is the **one-line prompt** for Task Description #3:
#"Generate a Python program for a Smart Healthcare Appointment Scheduling System that stores appointment ID, patient name, doctor name, appointment time, and consultation fee, uses Binary Search to find appointments by ID, and uses Merge Sort to sort appointments by time or consultation fee with proper comments and explanations."
# Smart Healthcare Appointment Scheduling System

# Appointment records
appointments = [
    {"id":101, "patient":"Ravi", "doctor":"Dr.Kumar", "time":10, "fee":500},
    {"id":102, "patient":"Anita", "doctor":"Dr.Rao", "time":12, "fee":300},
    {"id":103, "patient":"Sita", "doctor":"Dr.Mehta", "time":9, "fee":700},
    {"id":104, "patient":"Rahul", "doctor":"Dr.Sharma", "time":11, "fee":400}
]

# Binary Search to find appointment by ID
def binary_search(arr, target):
    left = 0
    right = len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid]["id"] == target:
            return arr[mid]
        elif arr[mid]["id"] < target:
            left = mid + 1
        else:
            right = mid - 1

    return "Appointment not found"


# Merge Sort to sort appointments
def merge_sort(arr, key):
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid], key)
    right = merge_sort(arr[mid:], key)

    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i][key] < right[j][key]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result


# Sort appointments by time
sorted_time = merge_sort(appointments, "time")
print("Sorted by Time:", sorted_time)

# Sort appointments by consultation fee
sorted_fee = merge_sort(appointments, "fee")
print("Sorted by Fee:", sorted_fee)

# Search appointment by ID
appointments_sorted_by_id = sorted(appointments, key=lambda x: x["id"])
result = binary_search(appointments_sorted_by_id, 103)
print("Search Result:", result)
```

OUPUT:

```
PS C:\Users\gerum\OneDrive\Desktop\AI_ASS_CODING> c:; cd "c:\Users\gerum\OneDrive\Desktop\AI_ASS_CODING" ; & "C:\Users\gerum\AppData\Local\Programs\Python\Python312\python.exe" "c:\Users\gerum\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher" "58781" "--" "C:\Users\gerum\OneDrive\Desktop\AI_ASS_CODING\ai_assistedcoding\lab_12.5" "c:\Users\gerum\.vscode\extensions\ms-python.debugpy-2025.18.0-win32
-x64\bundled\libs\debugpy\launcher" "58783" "--" "C:\Users\gerum\OneDrive\Desktop\AI_ASS_CODING\ai_assistedcoding\lab_12.5"
CODING\ai_assistedcoding\lab_12.5"
Sorted by Time: [{'id': 103, 'patient': 'Sita', 'doctor': 'Dr.Mehta', 'time': 9, 'fee': 700}, {'id': 101, 'patient': 'Ravi', 'doctor': 'Dr.Kumar', 'time': 10, 'fee': 5}sorted by Time: [{'id': 103, 'patient': 'Sita', 'doctor': 'Dr.Mehta', 'time': 9, 'fee': 700}, {'id': 101, 'patient': 'Ravi', 'doctor': 'Dr.Kumar', 'time': 10, 'fee': 500}, {'id': 104, 'patient': 'Rahul', 'doctor': 'Dr.Sharma', 'time': 11, 'fee':
 400}, {'id': 102, 'patient': 'Anita', 'doctor': 'Dr.Rao', 'time': 12, 'fee': 300}]
Sorted by Fee: [{'id': 102, 'patient': 'Anita', 'doctor': 'Dr.Rao', 'time': 12, 'fee': 300}, {'id': 104, 'patient': 'Rahul', 'doctor': 'Dr.Sharma', 'time': 11, 'fee': 400}, {'id': 101, 'patient': 'Ravi', 'doctor': 'Dr.Kumar', 'time': 10, 'fee': 500}, {'id': 103, 'patient': 'Sita', 'doctor': 'Dr.Mehta', 'time': 9, 'fee': 700}]
Search Result: {'id': 103, 'patient': 'Sita', 'doctor': 'Dr.Mehta', 'time': 9, 'fee': 700}
PS C:\Users\gerum\OneDrive\Desktop\AI_ASS_CODING>
```

Task 4:

Railway Ticket Reservation System

Scenario

A railway reservation system stores booking details such as ticket ID, passenger name, train number, seat number, and travel date. The system must:

1. Search tickets using ticket ID.
2. Sort bookings based on travel date or seat number.

   Student Task

   - Identify efficient algorithms using AI assistance.
   - Justify the algorithm choices.
   - Implement searching and sorting in Python.

PROMPT:

```
# **"Generate a Python program for a Railway Ticket Reservati
on System that stores ticket ID, passenger name, train numbe
r, seat number, and travel date, uses Binary Search to find t
ickets by ticket ID, and uses Merge Sort to sort bookings by
travel date or seat number with proper comments and documenta
tion."**
# Railway Ticket Records
```

CODE:

```python
# **"Generate a Python program for a Railway Ticket Reservation System that stores ticket ID, passenger name, train number, seat number, and travel date, uses Binary Search to find tickets by ticket ID, and uses Merge Sort to sort bookings by travel date or seat number with proper comments and documentation."**
# Railway Ticket Records
tickets = [
    {"id":201, "passenger":"Ramesh", "train":1201, "seat":15, "date":"2026-04-01"},
    {"id":202, "passenger":"Sita", "train":1205, "seat":10, "date":"2026-03-28"},
    {"id":203, "passenger":"Rahul", "train":1210, "seat":8, "date":"2026-04-05"},
    {"id":204, "passenger":"Anita", "train":1199, "seat":20, "date":"2026-03-30"}
]

# Binary Search for Ticket ID
def binary_search(arr, target):
    left = 0
    right = len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid]["id"] == target:
            return arr[mid]
        elif arr[mid]["id"] < target:
            left = mid + 1
        else:
            right = mid - 1

    return "Ticket not found"


# Merge Sort for sorting tickets
def merge_sort(arr, key):
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid], key)
    right = merge_sort(arr[mid:], key)

    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i][key] < right[j][key]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result


# Sort by travel date
sorted_date = merge_sort(tickets, "date")
print("Sorted by Travel Date:", sorted_date)

# Sort by seat number
sorted_seat = merge_sort(tickets, "seat")
print("Sorted by Seat Number:", sorted_seat)

# Search ticket by ID
tickets_sorted_by_id = sorted(tickets, key=lambda x: x["id"])
print("Search Result:", binary_search(tickets_sorted_by_id, 203))
```

OUPUT:

```
1launcher" "50701" "--" "C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING\ai_assistedcoding\lab_12.5"
Sorted by Time: [{'id': 103, 'patient': 'Sita', 'doctor': 'Dr.Mehta', 'time': 9, 'fee': 700}, {'id': 101, 'patient': 'Ravi', 'doctor': 'Dr.Kumar', 'time': 10, 'fee': 5Sorted by Time: [{'id': 103, 'patient': 'Sita', 'doctor': 'Dr.Mehta', 'time': 9, 'fee': 700}, {'id': 101
, 'patient': 'Ravi', 'doctor': 'Dr.Kumar', 'time': 10, 'fee': 500}, {'id': 104, 'patient': 'Rahul', 'doctor': 'Dr.Sharma', 'time': 11, 'fee': 400}, {'id': 102, 'patient': 'Anita', 'doctor': 'Dr.Rao', 'time': 12, 'fee': 300}]
Sorted by Fee: [{'id': 102, 'patient': 'Anita', 'doctor': 'Dr.Rao', 'time': 12, 'fee': 300}, {'id': 104, 'patient': 'Rahul', 'doctor': 'Dr.Sharma', 'time': 11, 'fee': 400}, {'id': 101, 'patient': 'Ravi', 'doctor': 'Dr.Kumar', 'time': 10, 'fee': 500}, {'id': 103, 'patient
': 'Sita', 'doctor': 'Dr.Mehta', 'time': 9, 'fee': 700}]
Search Result: {'id': 103, 'patient': 'Sita', 'doctor': 'Dr.Mehta', 'time': 9, 'fee': 700}
PS C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING>
```

Task 5:

Smart Hostel Room Allocation System
A hostel management system stores student room allocation details including student ID, room number, floor, and allocation date. The system needs to:

1. Search allocation details using student ID.
2. Sort records based on room number or allocation date.
   Student Task
   - Use AI to suggest optimized algorithms.
   - Justify the selections.
   - Implement the solution in Python.

PROMPT:

```
# Generate a Python program for a Smart Hostel Room Allocatio
n System that stores student ID, room number, floor, and allo
cation date, uses Binary Search to find records by student I
D, and uses Merge Sort to sort records by room number or allo
cation date with proper comments
# Hostel room allocation records
```

CODE:

```python
# Generate a Python program for a Smart Hostel Room Allocation System that stores student ID, room number, floor, and allocation date, uses Binary Search to find records by student ID, and uses Merge Sort to sort records by room number or allocation date with proper comments
# Hostel room allocation records
allocations = [
    {"student_id":101, "room":205, "floor":2, "date":"2026-03-01"},
    {"student_id":102, "room":101, "floor":1, "date":"2026-02-25"},
    {"student_id":103, "room":310, "floor":3, "date":"2026-03-05"},
    {"student_id":104, "room":150, "floor":1, "date":"2026-02-28"}
]

# Binary Search for Student ID
def binary_search(arr, target):
    left = 0
    right = len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid]["student_id"] == target:
            return arr[mid]
        elif arr[mid]["student_id"] < target:
            left = mid + 1
        else:
            right = mid - 1

    return "Student allocation not found"


# Merge Sort for sorting records
def merge_sort(arr, key):
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid], key)
    right = merge_sort(arr[mid:], key)

    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i][key] < right[j][key]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result


# Sort by room number
sorted_room = merge_sort(allocations, "room")
print("Sorted by Room Number:", sorted_room)

# Sort by allocation date
sorted_date = merge_sort(allocations, "date")
print("Sorted by Allocation Date:", sorted_date)

# Search allocation by student ID
allocations_sorted = sorted(allocations, key=lambda x: x["student_id"])
print("Search Result:", binary_search(allocations_sorted, 103))
```

OUTPUT:

```
PS C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING> cd; cd "c:\Users\perum\OneDrive\Desktop\AI_ASS_CODING"; & "C:\Users\perum\AppData\Local\Programs\Python\Python312\python.exe" "c:\Users\perum\.vscode\extensions\ms-python.debugpy-2025.10.0-win32-x64\bundled\libs\debugpy\launcher" "61992" "C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING\ai_assistedcoding\lab_12.5"
Sorted by Room Number: [{'student_id': 102, 'room': 101, 'floor': 1, 'date': '2026-02-25'}, {'student_id': 104, 'room': 150, 'floor': 1, 'date': '2026-02-28'}, {'student_id': 101, 'room': 205, 'floor': 2, 'date': '2026-03-01'}, {'student_id': 103, 'room': 310, 'floor': 3, 'date': '2026-03-05'}]
Sorted by Allocation Date: [{'student_id': 102, 'room': 101, 'floor': 1, 'date': '2026-02-25'}, {'student_id': 104, 'room': 150, 'floor': 1, 'date': '2026-02-28'}, {'student_id': 101, 'room': 205, 'floor': 2, 'date': '2026-03-01'}, {'student_id': 103, 'room': 310, 'floor': 3, 'date': '2026-03-05'}]
Search Result: {'student_id': 103, 'room': 310, 'floor': 3, 'date': '2026-03-05'}
PS C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING>
```

Task 6:

Online Movie Streaming Platform
A streaming service maintains movie records with movie ID, title, genre, rating, and release year. The platform needs to:

1. Search movies by movie ID.
2. Sort movies based on rating or release year.
   Student Task
   - Recommend searching and sorting algorithms using AI.
   - Justify the chosen algorithms.
   - Implement Python functions.

PROMPT:

```
# Generate a Python program for an Online Movie Streaming Pla
tform that stores movie ID, title, genre, rating, and release
year, uses Binary Search to find movies by movie ID, and uses
Merge Sort to sort movies by rating or release year with prop
er comments
# Movie records
```

CODE:

```python
# Generate a Python program for an Online Movie Streaming Platform that stores movie ID, title, genre, rating, and release year, uses Binary Search to find movies by movie ID, and uses Merge Sort to sort movies by rating or release year with proper comments
# Movie records
movies = [
    {"id":1, "title":"Inception", "genre":"Sci-Fi", "rating":8.8, "year":2010},
    {"id":2, "title":"Avatar", "genre":"Action", "rating":7.8, "year":2009},
    {"id":3, "title":"Interstellar", "genre":"Sci-Fi", "rating":8.6, "year":2014},
    {"id":4, "title":"Titanic", "genre":"Romance", "rating":7.9, "year":1997}
]

# Binary Search by Movie ID
def binary_search(arr, target):
    left = 0
    right = len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid]["id"] == target:
            return arr[mid]
        elif arr[mid]["id"] < target:
            left = mid + 1
        else:
            right = mid - 1

    return "Movie not found"


# Merge Sort for sorting movies
def merge_sort(arr, key):
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid], key)
    right = merge_sort(arr[mid:], key)

    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i][key] < right[j][key]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result


# Sort by rating
sorted_rating = merge_sort(movies, "rating")
print("Sorted by Rating:", sorted_rating)

# Sort by release year
sorted_year = merge_sort(movies, "year")
print("Sorted by Release Year:", sorted_year)

# Search movie by ID
movies_sorted = sorted(movies, key=lambda x: x["id"])
print("Search Result:", binary_search(movies_sorted, 3))
```

OUTPUT:

```
PS C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING>  c:; cd 'c:\Users\perum\OneDrive\Desktop\AI_ASS_CODING'; & 'C:\Users\perum\AppData\Local\Programs\Python\Python312\python.exe' 'c:\Users\perum\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\
launcher' '64802' '--' "C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING\ai_assistedcoding\lab_12.5"
Sorted by Rating: [{'id': 2, 'title': 'Avatar', 'genre': 'Action', 'rating': 7.8, 'year': 2009}, {'id': 4, 'title': 'Titanic', 'genre': 'Romance', 'rating': 7.9, 'year': 1997}, {'id': 3, 'title': 'Interstellar', 'genre': 'Sci-Fi', 'rating': 8.6, 'year': 2014}, {'id': 1, '
'title': 'Inception', 'genre': 'Sci-Fi', 'rating': 8.8, 'year': 2010}]
Sorted by Release Year: [{'id': 4, 'title': 'Titanic', 'genre': 'Romance', 'rating': 7.9, 'year': 1997}, {'id': 2, 'title': 'Avatar', 'genre': 'Action', 'rating': 7.8, 'year': 2009}, {'id': 1, 'title': 'Inception', 'genre': 'Sci-Fi', 'rating': 8.8, 'year': 2010}, {'id':
3, 'title': 'Interstellar', 'genre': 'Sci-Fi', 'rating': 8.6, 'year': 2014}]
Search Result: {'id': 3, 'title': 'Interstellar', 'genre': 'Sci-Fi', 'rating': 8.6, 'year': 2014}
PS C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING>
```

Task 7:

Smart Agriculture Crop Monitoring System
An agriculture monitoring system stores crop data with crop ID, crop name, soil moisture level, temperature, and yield estimate. Farmers need to:

1. Search crop details using crop ID.
2. Sort crops based on moisture level or yield estimate.
   Student Task
   - Use AI-assisted reasoning to select algorithms.
   - Justify algorithm suitability.
   - Implement searching and sorting in Python.

PROMPT:

```
# Generate a Python program for a Smart Agriculture Crop Moni
toring System that stores crop ID, crop name, soil moisture l
evel, temperature, and yield estimate, uses Binary Search to
find crops by crop ID, and uses Merge Sort to sort crops by m
oisture level or yield estimate with proper comments
# Crop data records
```

CODE:

```python
# Generate a Python program for a Smart Agriculture Crop Monitoring System that stores crop ID, crop name, soil moisture level, temperature, and yield estimate, uses Binary Search to find crops by crop ID, and uses Merge Sort to sort crops by moisture level or yield estimate with proper comments
# Crop data records
crops = [
    {"id":1, "name":"Wheat", "moisture":30, "temperature":25, "yield":200},
    {"id":2, "name":"Rice", "moisture":45, "temperature":28, "yield":350},
    {"id":3, "name":"Corn", "moisture":35, "temperature":27, "yield":300},
    {"id":4, "name":"Barley", "moisture":25, "temperature":22, "yield":180}
]

# Binary Search by Crop ID
def binary_search(arr, target):
    left = 0
    right = len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid]["id"] == target:
            return arr[mid]
        elif arr[mid]["id"] < target:
            left = mid + 1
        else:
            right = mid - 1

    return "Crop not found"


# Merge Sort for sorting crops
def merge_sort(arr, key):
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid], key)
    right = merge_sort(arr[mid:], key)

    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i][key] < right[j][key]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result

# Sort crops by moisture level
sorted_moisture = merge_sort(crops, "moisture")
print("Sorted by Moisture:", sorted_moisture)

# Sort crops by yield estimate
sorted_yield = merge_sort(crops, "yield")
print("Sorted by Yield:", sorted_yield)

# Search crop by ID
crops_sorted = sorted(crops, key=lambda x: x["id"])
print("Search Result:", binary_search(crops_sorted, 3))
```

OUTPUT:

```
PS C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING> c:; cd 'c:\Users\perum\OneDrive\Desktop\AI_ASS_CODING'; & 'C:\Users\perum\AppData\Local\Programs\Python\Python312\python.exe' 'c:\Users\perum\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '52249' '--' 'C:\Users\perum\OneDrive\Desktop\AI_ASS_CODI
tedcoding\lab_12.5'
Rice', 'moisture': 45, 'temperature': 28, 'yield': 350}]
Sorted by Yield: [{'id': 4, 'name': 'Barley', 'moisture': 25, 'temperature': 22, 'yield': 180}, {'id': 1, 'name': 'Wheat', 'moisture': 30, 'temperature': 25, 'yield': 200}, {'id': 3, 'name': 'Corn', 'moisture': 35, 'temperature': 27, 'yield': 300}, {'id': 2, 'name': 'Rice', 'moisture': 45, 'temperature': 28, 'yield': 350}]
Search Result: {'id': 3, 'name': 'Corn', 'moisture': 35, 'temperature': 27, 'yield': 300}
PS C:\Users\perum\OneDrive\Desktop\AI_ASS_CODING>
```

Task 8:

Airport Flight Management System
An airport system stores flight information including flight ID, airline name, departure time, arrival time, and status. The system must:

1. Search flight details using flight ID.
2. Sort flights based on departure time or arrival time.
   Student Task
   - Use AI to recommend algorithms.
   - Justify the algorithm selection.
   - Implement searching and sorting logic in Python.

PROMPT:

```
# Generate a Python program for an Airport Flight Management
System that stores flight ID, airline name, departure time, a
rrival time, and status, uses Binary Search to find flights b
y flight ID, and uses Merge Sort to sort flights by departure
time or arrival time with proper comments
# Flight records
```

CODE:

```python
'''
# Generate a Python program for an Airport Flight Management System that stores flight ID, airline name, departure time, arrival time, and status, uses Binary Search to find flights by flight ID, and uses Merge Sort to sort flights by departure time or arrival time with proper comments
# Flight records
flights = [
    {"id":101, "airline":"IndiGo", "departure":"10:30", "arrival":"12:45", "status":"On Time"},
    {"id":102, "airline":"Air India", "departure":"08:15", "arrival":"10:30", "status":"Delayed"},
    {"id":103, "airline":"SpiceJet", "departure":"14:00", "arrival":"16:20", "status":"On Time"},
    {"id":104, "airline":"Vistara", "departure":"09:45", "arrival":"11:50", "status":"Boarding"}
]

# Binary Search by Flight ID
def binary_search(arr, target):
    left = 0
    right = len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid]["id"] == target:
            return arr[mid]
        elif arr[mid]["id"] < target:
            left = mid + 1
        else:
            right = mid - 1

    return "Flight not found"


# Merge Sort for sorting flights
def merge_sort(arr, key):
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid], key)
    right = merge_sort(arr[mid:], key)

    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i][key] < right[j][key]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result


# Sort flights by departure time
sorted_departure = merge_sort(flights, "departure")
print("Sorted by Departure Time:", sorted_departure)

# Sort flights by arrival time
sorted_arrival = merge_sort(flights, "arrival")
print("Sorted by Arrival Time:", sorted_arrival)

# Search flight by ID
flights_sorted = sorted(flights, key=lambda x: x["id"])
print("Search Result:", binary_search(flights_sorted, 103))
```

OUTPUT: